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Engineering Lead - Geographic Information Systems (GIS)

Canada

TRENDING

JOB DESCRIPTION

Metrolinx is connecting communities across the Greater Golden Horseshoe. Metrolinx operates GO Transit and UP Express, as well as the PRESTO fare payment system. We are also building new and improved rapid transit, including GO Expansion, Light Rail Transit routes, and major expansions to Toronto's subway system, to get people where they need to go, better, faster and easier. Metrolinx is an agency of the Government of Ontario.

At Metrolinx, equity, diversity and inclusion are essential to living our values of serving with passion, thinking forward and playing as a team.

This role is an in-office position based at our 20 Bay Street location. Our team is highly collaborative, and much of our work benefits from in-person co-creation, relationship-building, and hands-on partnership with leaders across the organization.

The **Engineering & Safety Division** is hiring an **Engineering Lead, Geographic Information Systems (GIS)** whom provides strategic leadership for the development, governance, integration, and application of Geographic Information Systems across the Engineering and Safety Division (ESD). The Engineering Lead, GIS is

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responsible for ensuring that geospatial information and analysis are effectively integrated into business processes and decision-making across the organization.

Working at the intersection of engineering, asset management, capital planning, operations, and technology, this role establishes the strategic direction and governance framework for GIS while leading the development of advanced geospatial capabilities that support a common, system-level understanding of the railway network.

The role will enable the organization to integrate asset location with condition, age, performance, lifecycle, operational, and project information to support evidence-based decisions and optimize infrastructure investment.

Why This Role Matters

The Engineering Lead, GIS will play a critical role in enabling Metrolinx to make increasingly integrated, evidence-based decisions about its infrastructure and assets.

As the Asset Management Program continues to mature, the organization requires the ability to understand not only asset inventories, but also how asset condition, risk, performance, operational constraints, capital projects, and planned interventions interact spatially across the network.

This position provides the leadership required to establish GIS as a foundational component of that capability. Through effective governance, data integration, spatial analytics, and cross-functional leadership, the Engineering Lead, GIS will enable the organization to:

- Improve the prioritization of infrastructure investments.
- Coordinate capital interventions across multiple programs.

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- Identify spatial dependencies, conflicts, and opportunities.
- Strengthen risk-based asset management and State of Good Repair planning.
- Improve visibility of asset condition and lifecycle requirements.
- Integrate engineering, asset, operational, and project information.
- Support system-level planning across the railway network.
- Provide senior leaders with clear, spatially informed decision-support information.
- Establish consistent enterprise geospatial standards and practices.
- Advance the organization's digital asset management capabilities.

What Will I Be Doing?

Strategic GIS Leadership

- Establish and lead the strategic direction for GIS within the Engineering and Safety Division, ensuring alignment with enterprise Asset Management, State of Good Repair (SOGR), capital investment, and infrastructure planning objectives.
- Develop and maintain a division-wide GIS strategy, roadmap, and business plan that identifies priorities, capabilities, technology requirements, and opportunities for continuous improvement.
- Provide strategic advice and subject matter expertise to senior leadership on the use of geospatial information, analytics, and technology to support infrastructure and asset management decisions.
- Represent GIS interests in strategic planning, program development, technology initiatives, and organizational decision-making forums.
- Identify emerging geospatial technologies, analytical methods, and industry practices that can improve infrastructure planning, asset management, and operational performance.

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Geospatial Governance and Data Management

- Establish and govern division-wide geospatial standards, methodologies, data models, processes, and quality requirements.
- Provide leadership for geospatial data stewardship, ensuring that critical asset and infrastructure information is accurate, current, consistent, accessible, and fit for business use.
- Establish governance frameworks that clarify ownership, accountability, data stewardship, and standards across business units.
- Ensure alignment between GIS data and enterprise systems, including Enterprise Asset Management, engineering, operational, project, and capital planning systems.
- Develop and maintain a common geospatial framework for representing the railway network and its associated assets.

Asset Management and Spatial Analytics

- Lead the development and application of advanced geospatial analytics to support evidence-based asset management and investment decisions.
- Integrate spatial information with asset condition, age, performance, failure risk, maintenance history, operational constraints, and lifecycle information to support risk-based prioritization.
- Develop spatial analysis capabilities that identify infrastructure risks, dependencies, opportunities, and relationships that may not be apparent through conventional tabular analysis.
- Support the development of system-level State of Good Repair strategies through spatial analysis of asset condition, risk, criticality, and investment requirements.
- Enable the use of GIS to identify geographic concentrations of asset risk and support prioritization of infrastructure interventions.

Capital Planning and Program Coordination

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- Lead the use of geospatial analysis to coordinate capital interventions and work planning across multiple programs.
- Provide spatial analysis to identify conflicts, dependencies, access constraints, construction interfaces, and opportunities for coordinated delivery of capital and maintenance activities.
- Develop geospatial tools and decision-support products that enable project teams to understand the cumulative impacts of proposed interventions across the railway network.
- Support integrated planning between Engineering, Capital Projects Group (CPG), Operations, Asset Management, and other stakeholders.
- Enable spatially informed planning of infrastructure investments to minimize conflicts, optimize delivery opportunities, and improve the overall effectiveness of capital programs.

Risk and Decision Support

- Develop and implement geospatial approaches to infrastructure risk assessment and mitigation planning.
- Integrate spatial, engineering, operational, and asset data to support data-driven risk assessments.
- Provide senior leadership with geospatial analysis and visualization that clearly communicates infrastructure risk, investment requirements, project interactions, and potential service impacts.
- Develop dashboards, analytical models, maps, and other decision-support products that translate complex geospatial information into actionable business intelligence.
- Support scenario analysis to assess the potential impacts of infrastructure interventions, changes in asset condition, operational constraints, and capital investment strategies.

Cross-Functional Leadership

- Establish effective working relationships with Engineering, CPG, Operations, Asset Management,

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Technology, and other business areas to ensure GIS capabilities address enterprise priorities.
- Lead cross-functional initiatives involving multiple stakeholders, systems, datasets, and programs.
- Facilitate the development of common geospatial approaches and standards across organizational boundaries.
- Provide leadership and direction to GIS professionals, consultants, vendors, and project teams as required.
- Influence stakeholders and business partners to adopt consistent geospatial practices and leverage GIS as an enterprise decision-support capability.

Technology and Digital Asset Management

- Provide strategic direction for the evolution of GIS technology and its integration with enterprise digital asset management capabilities.
- Guide the integration of GIS with Enterprise Asset Management systems, engineering systems, operational data, capital planning systems, and other enterprise platforms.
- Assess technology solutions and architectural approaches to ensure GIS capabilities are scalable, secure, sustainable, and aligned with enterprise technology strategies.
- Support the development of digital representations of the railway network and infrastructure assets to enable improved analysis, planning, and decision-making.
- Promote automation, interoperability, and data integration to reduce duplication and improve the efficiency and reliability of geospatial workflows.

What Skills and Qualifications Do I Need?

- Completion of degree or diploma in Geographic Information Systems, Geography, Geomatics, Engineering, Computer Science, Data Analytics, or a related discipline.
- Demonstrates progressive experience in GIS, geospatial analytics, asset management,

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infrastructure planning, engineering, or a related field.

- Demonstrated experience leading GIS programs, enterprise geospatial initiatives, or complex cross-functional projects.
- Strong understanding of GIS principles, spatial databases, geospatial data management, spatial analytics, and enterprise GIS architecture.
- Experience applying GIS and spatial analytics to infrastructure, transportation, utilities, asset management, capital planning, or other complex physical networks.
- Demonstrated experience integrating GIS with enterprise systems and diverse business data sources.
- Experience developing geospatial governance frameworks, standards, data models, and data stewardship practices.
- Experience using spatial analysis to support risk assessment, asset prioritization, investment planning, or operational decision-making.
- Demonstrated experience working with senior stakeholders and leading initiatives across multiple organizational functions.
- Strong understanding of asset management principles and the application of data and analytics to support lifecycle planning and State of Good Repair.
- Experience leading consultants, technical specialists, vendors, and/or multidisciplinary project teams would be considered an asset.
- Professional certification in GIS, Geomatics, Engineering, Asset Management, or a related discipline would be considered an asset.
- Leadership Competencies: The successful candidate will demonstrate:
    - Strategic Leadership: Ability to establish a long-term vision for GIS and translate organizational priorities into actionable strategies and programs.
    - Systems Thinking: Ability to understand relationships between assets, infrastructure, projects, operations, risks, and business processes across a complex transportation network.
    - Analytical Leadership: Ability to transform complex spatial and non-spatial data into

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meaningful insights that support investment and operational decisions.
- Collaboration and Influence: Ability to build consensus and influence stakeholders across organizational boundaries without relying solely on formal authority.
- Business Acumen: Understanding of how GIS, asset management, capital planning, engineering, and operations intersect to support organizational objectives.
- Innovation: Ability to identify and implement new technologies, analytical approaches, and business processes that improve organizational performance.
- Communication: Ability to communicate complex technical and analytical concepts clearly to technical specialists, business leaders, and executives.

Don’t Meet Every Requirement?

If you’re excited about working with Metrolinx but your past experience doesn’t quite align with every qualification of this posting, we encourage you to apply. You just might be the right candidate for this or other roles. We are always looking for great talent to join our team.

We invite all interested individuals to apply and encourage applications from members of equity-deserving communities, including those who identify as Indigenous, Black, racialized, women, people with disabilities, and people with diverse gender identities, expressions and sexual orientations.

Accommodation:
Metrolinx is an equal opportunity employer and committed to a diverse and inclusive workforce. We are also committed to offering reasonable accommodation to job applicants in accordance with applicable legislation. If you require assistance or an accommodation at any point during the hiring process, please contact us at: 416-202-5601 or email hr.recruitment@metrolinx.com.

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Application Process:

All applicants must be legally entitled to work in Canada. Metrolinx will be using email to communicate with you for all job competitions. It is your responsibility to include an updated email address that is checked daily and accepts emails from unknown users. As we send time-sensitive correspondence, we recommend that you check your email regularly. If no response is received, we will assume you are no longer interested in pursuing the opportunity. Please be advised that a Criminal Record Check may be required of the successful candidate.

Should it be determined that any background information provided is misleading, inaccurate or incorrect, Metrolinx reserves the right to discontinue with the consideration of your application.

We thank all applicants for their interest, however, only those selected for further consideration will be contacted.

WE ARE AN EQUITABLE AND INCLUSIVE EMPLOYER.

#Li-MM3

APPLY NOW

JOB INFO

Job Identification	117344
Job Category	Management
Posting Date	10/09/2026, 02:12 PM
Apply Before	10/26/2026, 11:59 PM
Job Schedule	Full Time
Locations	20 Bay Street
Is this a Contract Position?	No
Minimum Pay Range	$110,854.00
Maximum Pay Range	$153,872.00
Division	Engineering & Safety

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